

UCS503

SmartToken

Planning Presentation

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Introduction

Overview

SmartToken is a system designed to help faculty and teaching assistants record student participation during live university labs quickly and with minimal disruption to the teaching process.

Importance

In a typical lab setting, pausing to record participation can interrupt the flow of teaching. This often leads to missed awards, delayed entries, or incorrect attribution. SmartToken aims to address these issues by making the process faster and more reliable, including during network disruptions.

Presentation Objectives

- Outline the current problem and lab workflow
- Present the proposed SmartToken solution and user roles
- Describe system design, offline support, and synchronization
- Share the development plan, evaluation approach, and project timeline

Project Scope

Project Aims

Recording participation awards during a live university lab interrupts faculty workflow and can result in missed, delayed, or incorrectly attributed entries. SmartToken seeks to reduce this interruption while preserving faculty control, maintaining correction history, and ensuring reliable operation during network failure.

Deliverables

- Faculty and TA controlled web/PWA system
- Configurable and versioned token rules
- **Offline operation** with local storage
- Conflict-safe synchronization
- Append-only audit history for corrections
- **Private student view portal**
- Deployment in live university labs

Functionality

Faculty or TA selects a student or group and records a participation award. The system stores the event securely, supports offline use, synchronizes when connectivity returns, retains a full correction history, and allows each student to view only their own record.

Proposed Functions

Fast Participation Entry

Enable quick selection of a student or group and recording of an award with minimal interruption. Selection method being a frame where names of all students are written in alphabetic order in boxes

Configurable Token Rules

Token rules are configurable, versioned, and historically traceable. Each award remains linked to the specific rule version under which it was granted.

Offline Operation

Faculty can continue recording even when internet connectivity is unavailable. Events are stored locally on the device and synchronized once the network is restored.

Append-Only Corrections

Corrections do not erase prior history. Both the original event and the subsequent correction remain visible for audit purposes.

Private Student View

Students can access their own token history and relevant corrections. Access is restricted so that no student can view another student's records.

Target Users

Faculty

Primary user responsible for conducting the lab and recording participation awards.

Student

Views their own token history and any related corrections through a private portal.

Administrator

Manages system configuration, access control, and related administrative functions.

System Features

Quick Student Selection

Supports search and recent/frequent lists so that faculty can **locate a student efficiently** during a live session.

Local Offline Storage

Events are retained on the device when network connectivity is unavailable, ensuring recording is not interrupted.

Participation Event Recording

Each award is stored with student identity, award value, recorder, timestamp, and the applicable rule version.

Interfaces

User Interface

Faculty opens SmartToken on a phone or laptop, locates the student via search or a recent list, selects the award value, and records the entry. The interaction is designed to require only a few seconds. Students access a private portal to view their personal history. Selection methods will be evaluated through a small timing study to determine the most effective approach in a live lab setting.

Hardware & Software Interfaces

Implemented as a web application / Progressive Web App accessible via laptop and mobile browsers. Local storage supports offline recording. AI assistance (such as transcription) is available only when explicitly initiated by faculty and is not required for core operation.

Communication Interfaces

Network connectivity is used for synchronization when available. Recording continues during network outages. The system **does not perform continuous classroom recording or listening.**

Performance Goals

Requirements

Token entry should be **sufficiently fast to avoid interrupting teaching**. The system must remain functional during network outages and synchronize correctly afterward. Attribution errors should be minimized, and corrections must preserve full history. These aspects will be evaluated in actual university laboratory settings.

Importance

If the recording process is slow or fails when connectivity is lost, users are likely to revert to manual methods or omit awards. Meeting these performance goals is essential for practical adoption and for meaningful comparison against the existing manual process.

Achievement Strategies

Developed as a Progressive Web App with robust local storage. Lightweight interface design combined with measured interaction testing. Conflict-aware synchronization. Evaluation conducted through supervised live-lab deployment.

Security Measures

Security Overview

The system is designed to protect **participation data and uphold user privacy**. Faculty retain **full control over awards**. **Students are limited to viewing** their own history. Continuous classroom recording, continuous listening, speaker identification, automatic grading, and automatic token awards are explicitly outside the scope of the project.

Access Control

Students can't access other students data, they can view only theirs. That option only resides with teacher

9 · GANTT CHART

Project Timeline — Gantt Chart

■ Task ■ Milestone

Week →

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17



Conclusion

SmartToken is a **faculty-controlled** web/PWA system that enables faculty and TAs to record participation awards efficiently during live labs. It supports configurable rules, offline operation, safe synchronization, correction history, and private student access. The core problem addressed is the interruption of teaching workflow and the associated risks of missed, delayed, or incorrectly attributed records.

The goal is to provide a **fair and consistent** way to recognize student participation while **assisting** faculty during live labs. The system maintains a reliable **history of participation** and corrections without replacing faculty judgment.